NATIONAL UNIVERSITY FAIRVIEW

School of Engineering and Technology • Computer Engineering Department

LAB ACTIVITY No. 8: "Testing and Form Validation"

CPSOFT30L – Software Design Laboratory

Field	Details
Project Title	GuideHub – NU Fairview Guidance and Counseling Center Management System
Submitted To	Engr. Benedict Zurbito (CpE Professor)
Submitted By	Dela Cruz, Felonia, Jardin, Laberinto, Ongpauco
Date Submitted	October 13, 2026

Activity 1 – Review the SRS (Functions & Form Validation Rules Tested)

- **1. Sign-Up Wizard (auth/signup.html & js/auth.js):** 3-step institutional registration with role selection (Student @students.nu-fairview.edu.ph vs. Faculty @nu-fairview.edu.ph), dynamic SHS/College year level dropdowns, Student/Employee ID format regex validation (YYYY-XXXXXX / FAC-XXXX), password confirmation matching, and live 6-digit Gmail SMTP OTP verification.
- **2. Login Form (auth/login.html & js/auth.js):** Inline institutional email username prefix input + domain selector dropdown (@students.nu-fairview.edu.ph / @nu-fairview.edu.ph), HTML5 required field validation, password visibility toggle, and role-based portal redirection.
- **3. Main Dashboard (portals/student.html, counselor.html, faculty.html, head.html, admin.html):** Role-based access control (requireRole), tab navigation across Appointments, Walk-In Digital Triage, Referrals, Confidential Case Notes, and Analytics, plus real-time Firestore synchronization.
- **4. Profile Dashboard & User Identity Display:** Navigation bar identity badge (Full Legal Name, Student/Employee ID, Role Badge), pre-populated student profile details in consultation/walk-in forms, and Admin role management.
- **5. Session Logout & Browser Back-Button Lock:** Sign Out clears active session and Firebase Auth state; window.location.replace, popstate/pageshow listeners, and Cache-Control: no-store headers prevent Back-button access across authentication boundaries.

Activity 2 – Identified Test Scenarios

A. Sign-Up Scenarios

- **Valid Registration:** Complete 3-step wizard with valid NU Fairview details and valid 6-digit OTP code.
- **Empty Required Fields:** Leaving Program, Year Level, Student ID, Full Name, Email Prefix, or Password blank.
- **Invalid Email / Restricted Domain:** Entering spaces, invalid symbols, or under-3-character username prefixes.
- **Invalid Student/Employee ID Format:** Entering letters in Student ID or not following YYYY-XXXXXX / FAC-XXXX.
- **Invalid Password & Password Mismatch:** Entering fewer than 6 characters or mismatched Password and Confirm Password.

- **Duplicate Account / ID:** Attempting to register an already existing institutional email or Student/Employee ID.

B. Login Scenarios

- **Valid Credentials:** Logging in with valid username prefix + institutional domain dropdown + valid password.
- **Invalid Username/Email or Password:** Entering an unregistered username prefix or wrong password.
- **Empty Username / Empty Password / Empty Form:** Submitting the login form with one or all required fields blank.

C. Main Dashboard Scenarios

- **Successful Access & Data Display:** Authenticated user lands on assigned portal and views live appointments, walk-in queue, and metrics.
- **Navigation Links & Interactive Buttons:** Switching dashboard tabs, booking a consultation, and generating a walk-in queue ticket (W-101).
- **Unauthorized Access & Back-Button Guard:** Blocking unauthenticated/wrong-role access and locking Browser Back button after login and logout.

D. Profile Dashboard Scenarios

- **Profile Information Display:** Displaying accurate Full Legal Name, Student/Employee ID, Role Badge, and Program/Department.
- **Profile Role Update (Admin Control):** Updating a user role in the Admin Portal and verifying Firestore users/{role} update and audit log.

Activity 3, 5, 6, 7 & 8 – Complete Test Case Documentation (18 Test Cases)

ID	Test Scenario	Input / Condition	Expected Result	Actual Result	Status
TC-001	Valid Sign-up (Student)	Role: Student Program: BS Computer Engineering Year: 2nd Year (College) ID: 2024-1039006 Name: Jose Villar Jardin Email: jardinjv@students.nu-fairview.edu.ph Pass: Password123! + Valid 6-digit OTP	6-digit OTP is dispatched via Gmail SMTP; upon OTP entry, account usr_student_jose_j is saved under users/students and redirected to Student Portal.	OTP received in Outlook inbox; account created under users/students and redirected to portals/student.html.	
TC-002	Empty Sign-up Required Field	Leave Full Legal Name or Student ID Number blank and click Continue / Send 6-Digit OTP.	System blocks progression and displays warning toast & inline error banner ("Please enter your Full Legal Name" / "Please enter your Student ID Number").	Progression blocked; warning toast and red error alert displayed on screen.	
TC-003	Invalid Email Username / Format	Enter invalid username prefix ("ab" or "user@gmail.com") in registerEmailPrefix.	System strips @ domain suffix automatically or blocks submission with	@ suffix stripped automatically; short/invalid prefix	

ID	Test Scenario	Input / Condition	Expected Result	Actual Result	Status
			"Invalid institutional email username. Must be at least 3 characters."	rejected with inline error message.	
TC-004	Invalid Student ID Format	Enter "12345" or letters "ABCD-123" in Student ID Number on Step 2.	Non-numeric characters are blocked on keypress; invalid pattern displays "Invalid Student ID format. Expected format: YYYY-XXXXXX".	Letters blocked from input; "12345" rejected with expected format error toast.	
TC-005	Invalid Password (Too Short)	Enter "123" (less than 6 characters) in Password and Confirm Password.	Form blocks OTP dispatch and displays "Your password must contain at least 6 characters."	Submission blocked; inline error banner and warning toast displayed.	
TC-006	Password Mismatch on Sign-up	Password: Password123! Confirm Password: Password999!	System blocks OTP dispatch and displays "The passwords you entered do not match. Please verify and re-type."	Error alert banner "Password Mismatch" and error toast displayed; OTP not sent.	
TC-007	Duplicate Email / ID on Sign-up	Email: jardinjv@students.nu-fairview.edu.ph or ID: 2024-1039005 (Already registered).	Server returns HTTP 409 / Step 2 blocks ID and displays "This institutional email / ID is already registered."	Blocked at validation check with message "This institutional email is already registered."	
TC-008	Invalid 6-Digit OTP Code	Enter "000000" (incorrect 6-digit OTP) on Step 4 verification screen.	System rejects code, decrements remaining attempts ("Incorrect verification code. 2 attempt(s) remaining."), and clears OTP boxes.	Error toast displayed with remaining attempts count; account creation prevented.	
TC-009	Valid Login (Student & Staff Domains)	Prefix: jardinjv Domain: @students.nu-fairview.edu.ph Password: 123456	User is authenticated, welcome toast appears, and browser redirects to portals/student.html.	Authenticated as Jose Villar Jardin; redirected to portals/student.html.	
TC-010	Invalid Login Credentials	Prefix: jardinjv Domain: @students.nu-fairview.edu.ph Password: WrongPass123	Authentication fails and displays error toast: "Invalid institutional email or password. Please verify your credentials."	Login rejected; error toast displayed and user remains on auth/login.html.	
TC-011	Empty Login Fields	Leave email-prefix and/or login-password blank and click Sign In to Workspace.	HTML5 required attribute prevents form submission and prompts user to fill out the required field.	Browser displays native required field validation tooltip; form does not submit.	
TC-012	Main Dashboard	Log in as mark.d@nu-	Main Dashboard loads	Dashboard loaded cleanly;	

ID	Test Scenario	Input / Condition	Expected Result	Actual Result	Status
	Access & Data Load	fairview.edu.ph (Counselor) or laberintogc@students.nu-fairview.edu.ph (Student).	without console errors; appointments, walk-in queue, and metrics cards populate from Firestore.	live appointments and status badges rendered accurately.	
TC-013	Dashboard Navigation & Booking	Click navigation tabs (Book Consultation, Walk-In Queue, My Appointments) and submit a booking.	Selected section displays smoothly; submitting valid booking creates apt_* document in Firestore and updates list live.	Tab views switched properly; new appointment saved and appeared immediately in table.	
TC-014	Profile Information Display	Log in as jardinjv@students.nu-fairview.edu.ph and inspect Navbar & Booking Profile card.	Displays Full Name (Jose Villar Jardin), ID (2024-1039005), Role Badge (Student), and Program (BS Computer Engineering).	All profile fields, ID number, role badge, and program displayed accurately.	
TC-015	Profile Role Update (Admin Portal)	Log in as Admin (admin.alex@nu-fairview.edu.ph), search user in User Directory, and change role selector.	User profile moves to target users/{role} document in Firestore, audit log is recorded, and confirmation toast is shown.	Role updated in Firestore users/{role}, toast displayed, and ROLE_UPDATED logged.	
TC-016	Unauthorized Portal Access Guard	While logged in as Student (or logged out), manually navigate to /portals/admin.html.	System blocks access, displays "Access restricted. Redirecting to your assigned workspace." (or redirects to Login if unauthenticated).	Access blocked immediately; redirected back to portals/student.html or auth/login.html.	
TC-017	Logout Functionality	Click Sign Out button in the top navigation bar.	Session storage and Firebase Auth session are cleared; user is redirected to auth/login.html.	Session cleared and redirected to auth/login.html.	
TC-018	Browser Back-Button Lock (Post-Login & Post-Logout)	1. Log in to Portal, then press Browser Back button. 2. Click Sign Out, then press Browser Back button.	1. While logged in, pressing Back stays on Portal (does not return to Login). 2. After logout, pressing Back stays on Login (does not reopen Portal).	History locked via location.replace and popstate/pageshow guards; Back button blocked in both states.	

Activity 9 & 10 – Defect Documentation & Retesting Results

Defect ID	Test Case	Problem	Expected	Actual (Before Fix)	Action Taken	Retest Status
D-001	TC-001 / TC-007	Uniqueness verification failed during signup when checking legacy	Clean uniqueness check across users/{role} and immediate OTP	Sign-up threw error on Step 3 and blocked OTP email dispatch.	Refactored findAccountByEmail and findAccountById in server/db.js to read	

Defect ID	Test Case	Problem	Expected	Actual (Before Fix)	Action Taken	Retest Status
		collections ("Failed to verify account uniqueness").	dispatch.		users/{role} and auto-clean orphaned Auth records.	
D-002	TC-001	Clicking "Resend Verification Code" on Step 4 failed with "All fields are required".	New 6-digit OTP email is sent and 5-minute timer resets.	btnResendOtp called sendInstitutionalOTP without passing pendingProfileData (role, idNumber), causing HTTP 400.	Updated btnResendOtp handler in auth/signup.html to pass pendingProfileData payload.	
D-003	TC-001 / TC-014	User documents were duplicated in root users and nested users/students/users subcollections, causing conflicts when re-creating deleted test accounts.	Each user is stored strictly once under users/{role} (usr_[role]_[name]).	Multiple copies existed in Firestore and orphaned records remained in Firebase Auth.	Removed duplicate root/subcollection writes in js/auth.js, js/services/db.js, and scripts/seed-accounts.js; purged duplicate docs.	
D-004	TC-017 / TC-018	Pressing browser Back button after logout displayed cached dashboard, or pressing Back after login returned to login form.	Browser Back button should not cross authentication boundaries without re-authenticating.	window.location.href left previous pages in browser history stack and bfcache.	Replaced location.href with window.location.replace, added popstate/pageshow history locks in js/auth.js, and added Cache-Control: no-store headers.	

Activity 11 – Final End-to-End Regression Testing

Final regression testing was executed across the complete user lifecycle: Sign-Up (Step 1 to Step 4 OTP Verification) → Login (with Institutional Domain Dropdown) → Main Dashboard (Appointments & Walk-In Queue) → Profile Display → Logout (with Browser Back-Button Lock). All 18 test cases passed without regression errors.

Guide Questions & Answers

1. Why is software testing important before deployment?

Software testing ensures that the web application meets all functional, security, and usability requirements before real users access it. In a sensitive system like GuideHub (which handles confidential student counseling records under RA 9258 and RA 10173), testing prevents critical defects such as unauthorized role access, broken OTP verification, or duplicate database records from occurring in production.

2. What is the difference between testing and debugging?

Testing is the process of executing the application with valid and invalid inputs to verify behavior and discover whether defects exist (e.g., finding that clicking "Resend OTP" returns an error). Debugging is the developer activity performed after

a test fails to trace the root cause in the source code (e.g., inspecting auth/signup.html and server/server.js) and fix the underlying logic.

3. What is the purpose of form validation?

Form validation ensures that user input is complete, accurately formatted, and safe before it is processed or stored in the database. In GuideHub, form validation enforces NU Fairview institutional email domains (@students.nu-fairview.edu.ph and @nu-fairview.edu.ph), validates Student/Employee ID formats (YYYY-XXXXXX), prevents password mismatches, and blocks empty or malformed submissions.

4. What is the difference between positive and negative testing?

Positive testing verifies that the system works as expected when provided with valid, normal inputs (e.g., signing up with a valid NU Fairview email, valid Student ID, and correct 6-digit OTP). Negative testing verifies that the system gracefully rejects invalid, unexpected, or missing inputs with clear error messages without crashing (e.g., entering an invalid ID format, mismatched passwords, wrong OTP, or blank fields).

5. Why should test cases be based on the SRS?

The Software Requirements Specification (SRS) defines the exact functional and security expectations agreed upon for the project. Designing test cases directly from the SRS guarantees that every required feature—such as role-based dashboards, institutional email gating, and appointment triage—is verified against objective criteria rather than guesswork.

6. Why should both expected and actual results be recorded?

Recording both the Expected Result (defined before running the test based on the SRS) and the Actual Result (observed during test execution) allows testers to objectively compare behavior, immediately spot discrepancies (PASS vs. FAIL), and maintain a clear audit trail for debugging and retesting.

7. What should be done when a test case fails?

When a test case fails, the tester must: (1) mark the test case status as FAIL, (2) record the exact actual behavior and capture screenshots/logs in the Defect Table, (3) trace and fix the root cause in the code, and (4) perform retesting and regression testing to confirm the fix works.

8. Why is retesting necessary after correcting a defect?

Retesting (confirmation testing) verifies that the code change actually resolved the reported defect under the exact same test conditions, while regression testing ensures that the fix did not accidentally break other working features (such as login, dashboard navigation, or Firestore synchronization).

9. What problems did your group discover during testing?

During testing, our group discovered four main defects: (1) an account uniqueness check error during sign-up when querying across legacy collections, (2) a missing payload parameter (pendingProfileData) on the "Resend Verification Code" button, (3) duplicate user documents inside nested users > students > users subcollections that prevented re-registering deleted test accounts, and (4) browser Back-button navigation allowing users to view cached pages after login/logout.

10. How did testing improve your web application?

Systematic testing made GuideHub significantly more reliable, secure, and presentation-ready. It streamlined our Firestore database schema into clean role documents (users/students, users/counselors, etc.), ensured 100% reliable 6-digit OTP email delivery and resend functionality, enforced strict role-based portal redirection, and locked browser Back-button history across login and logout sessions.

Group Reflection

Through Lab Activity No. 8, our group realized that building features is only half of software engineering—systematic testing and form validation are what turn a prototype into a dependable application. By testing both positive and

negative scenarios across our Sign-Up Wizard, Institutional Domain Login, Role Dashboards, and Profile views, we caught subtle edge cases (such as the Resend OTP payload bug, Firestore document duplication, and browser Back-button caching) that we would have missed during casual use. Following the structured cycle of Test → Document Defect → Fix Code → Retest gave us full confidence that GuideHub satisfies our SRS and is ready for live demonstration.

Evaluation Rubric

Criteria	Percentage	Score
Test Case Design	20%	15
Functional Testing	20%	10
Form Validation	20%	5
Defect Identification and Correction	15%	10
Retesting and Verification	10%	5
Test Documentation	10%	5
Screenshots and Evidence	5%	0
Total	100%	50